Stochastic Quantization: Interpolating Between Underdamped and Overdamped Langevin Dynamics

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Motivation: The Stochastic Quantization Program

Goal:

Sample from a Euclidean quantum field theory described via a target measure

$$d\mu[\varphi] = \frac{1}{Z} e^{-S_E[\varphi]}\, \mathcal{D}\varphi.$$

Challenge:

Direct sampling from μ is highly intractable.

Strategy: Stochastic Quantization (Parisi and Wu, 1981)

Construct stochastic Langevin dynamics ϕ_τ: We introduce a fictitious Langevin time τ and evolve the field $\phi(x,\tau)$ via a stochastic partial differential equation:

$$\frac{\partial \phi(x,\tau)}{\partial \tau} = -\frac{\delta S_E[\phi]}{\delta \phi(x,\tau)} + \xi(x,\tau)$$

where ξ is space fictitious-time white noise.

Motivation: The Stochastic Quantization Program

We require the solution $\phi(x, \tau)$ of the Langevin equation,

$$\frac{\partial \phi(x, \tau)}{\partial \tau} = -\frac{\delta S_E[\phi]}{\delta \phi(x, \tau)} + \xi(x, \tau)$$

to satisfy the following properties:

- **Well-Definedness:** The solution ϕ must exist and be globally well-defined. (Mourrat and Weber, 2017)
- **Unique Invariance:** The target measure $\mu[\phi]$ is the unique equilibrium distribution. (Tsatsoulis and Weber, 2018)
- **Ergodicity:** The probability law of the field converges to the Euclidean measure:
$$P(\phi_\tau) \xrightarrow{\tau \to \infty} \mu[\varphi]$$

$\Rightarrow$ *Compute n-point correlation functions by taking ergodic averages of ϕ_τ after a sufficient burn-in time.*

Background: The Φ^4_2 Model on the Torus

The target Euclidean quantum field theory on the torus $\mathbb{T}^2$ is formally defined by the path integral measure:

$$\mu[\varphi] = \frac{1}{Z} e^{-S_E[\varphi]} \mathcal{D}\varphi$$

where the Euclidean action is given by:

$$S_E[\varphi] = \int_{\mathbb{T}^2} \left(\frac{1}{2} |\nabla \varphi|^2 + \frac{1}{2} m^2 \varphi^2 + \frac{\lambda}{4!} \varphi^4 \right) d^2 x$$

The Mathematical Pathology (UV Divergence):

- The free theory is described by a Gaussian random field Φ with covariance operator $C = (-\Delta + m^2)^{-1}$.
- Consider the field $\Phi_N = P_N \Phi$ projected onto the first N Fourier modes. Taking the expectation over the i.i.d. Gaussian coefficients, the pointwise variance is:

$$C_N(x, x) = \mathbb{E}[\Phi_N(x)^2] = \sum_{|n| \leq N} \frac{1}{|n|^2 + m^2} \sim \log N \xrightarrow{N \to \infty} \infty$$

- It follows that Φ is a distribution and the non-linear interaction φ^4 is ill-defined.

Solution: Wick Renormalization

To obtain a fully rigorous model, we apply Wick Renormalization by replacing the interaction: $\varphi^4 \to :\varphi^4:$.
This subtracts the diverging variance $C = \mathbb{E}[\varphi^2]$:

$$:\varphi^4: = \varphi^4 - 6C\varphi^2 + 3C^2$$

The continuum path integral is mathematically well-defined, requiring no momentum cutoff:

$$d\mu[\varphi] = \frac{1}{Z} \exp\left(-\int_{\mathbb{T}^2} \left[\frac{1}{2}|\nabla\varphi|^2 + \frac{1}{2}m^2\varphi^2 + \frac{\lambda}{4!} :\varphi^4:\right] d^2x\right) \mathcal{D}\phi$$

Computational Significance

Consequently, this continuum model can be simulated by discretizing onto a lattice and taking the limit of vanishing grid spacing $a \to 0$.

The ULD Quantization of the $O(N)$ Φ^4_2-Model

Goal: Sample from an N-component Φ^4-theory described by the Euclidean action:

$$S_E[\varphi_N] = \int_{\mathbb{T}^2} \left[\frac{1}{2} \sum_{j=1}^{N} \left(|\nabla \varphi_{N,j}|^2 + \varphi_{N,j}^2 \right) + \frac{1}{4N} : \left(\sum_{j=1}^{N} \varphi_{N,j}^2 \right)^2 : \right] d^2x$$

We introduce a fictitious inertia $\varepsilon^2 > 0$ to construct the underdamped Langevin dynamics:

$$\frac{\partial \phi_{N,j}}{\partial \tau} = \pi_{N,j}$$

$$\varepsilon^2 \frac{\partial \pi_{N,j}}{\partial \tau} = -\pi_{N,j} - \frac{\delta S_E[\phi_N]}{\delta \phi_{N,j}} + \sqrt{2}\xi_{N,j}$$

Research Questions:

- **Well-Definedness & Ergodicity:** Does the ULD system sample from the target marginal measure $\mu[\phi_N]$? (Gubinelli, et al. 2022, Tolomeo, 2023 and Liu et al., 2025)
- **The Mean-Field Limit:** What is the asymptotic behavior as $N \to \infty$? (Liu et al., 2025 and Czernik, In Preparation)

The Smoluchowski Kramers Approximation I

Research Question:

Do we recover the OLD dynamics as the inertia $\varepsilon \to 0$? (Czernik, Liu, Liu, 2026)

Physical Intuition

As $\varepsilon \to 0$, the system loses its inertia: The damped wave propagation collapses into heat diffusion.

Mathematical Challenges

- **Loss of Smoothing:** The heat operator $e^{(\Delta-1)\tau}$ smooths out the fluctuations of the noise ξ; the damped wave operator not: it merely conserves it.
- **Global Stability:** Because the noise constantly adds energy into the system, we cannot argue with energy conservation!

  $\Rightarrow$ *controlling ϕ from blowing up is difficult.*
- **Uniformity:** To take the limit, all stability bounds must be independent of ε.

Main Result: The Overdamped Limit (Czernik, Liu, and Liu, 2026)

For any inertia $\varepsilon > 0$, the equilibrium distribution of the dynamic trajectories (ϕ_N, π_N) factorizes into the target path integral:

$$d\mu_\varepsilon[\varphi_N, \varpi_N] = \frac{1}{Z_\varepsilon} e^{-S_E[\varphi_N]} e^{-\frac{\varepsilon^2}{2}\int_{\mathbb{T}^2}\sum_{j=1}^N \varpi_{N,j}^2\, d^2x} \mathcal{D}\varphi_N \mathcal{D}\varpi_N$$

As $\varepsilon \to 0$, the momentum $\pi_N \to 0$, and the spatial field trajectories ϕ_N recover the standard Overdamped Langevin Dynamics.

Proof Strategy: Uniform Energy Bounds

To guarantee the convergence, we control the Hamiltonian uniformly in $\varepsilon \in (0, 1]$:

$$\sup_{\varepsilon\in(0,1]} \int_{\mathbb{T}^2} \left[\frac{1}{2}\sum_{j=1}^N \left(|\nabla\phi_{N,j}|^2 + \phi_{N,j}^2\right) + \frac{1}{4N} :\left(\sum_{j=1}^N \phi_{N,j}^2\right)^2: + \frac{\varepsilon^2}{2}\sum_{j=1}^N \pi_{N,j}^2 \right] d^2x < C \quad \text{a.s.}$$

Computational Significance: The Tuning Parameter ε

The $\varepsilon \to 0$ convergence mathematically justifies treating the fictitious inertia ε as a tuning parameter if applied to MCMC algorithms, as the invariant target measure $\mu[\varphi_N]$ is preserved.

The Algorithmic Trade-off (Batrouni et al., 1985, Duane et al. 1987, Neal 2011)
Tuning ε allows us to balance the sampling dynamics:

Overdamped ($\varepsilon \to 0$)	**Underdamped ($\varepsilon > 0$)**
Diffusive exploration: poor mixing & critical slowing down	Ballistic exploration: better mixing & faster decorrelation times
Stable numerical integration	Oscillations lead to instability

Summary:

The following diagram summarizes the relationships between the N-component models, their mean-field limits, and the respective Smoluchowski Kramers Approximations.

$$
\begin{tikzcd}
\text{N-Component ULD} \arrow[rr, "N\to\infty", "\text{[Liu et al., 2025]}"'] \arrow[d, "\varepsilon\to 0"'] & & \text{Mean-Field ULD} \arrow[d, "\varepsilon\to 0"] \\
\text{N-Component OLD} \arrow[rr, "\text{[Shen et al., 2022]}", "N\to\infty"'] & & \text{Mean-Field OLD}
\end{tikzcd}
$$

Figure: Diagram of limits between the underdamped and overdamped Langevin models.

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